

Ideation Phase

Brainstorm & Idea Prioritization Template

Date	15 July 2026
Team ID	SWTID-2026-5009
Project Name	HouseHunt – Smart House Rental Platform
Maximum Marks	4 Marks

Brainstorm & Idea Prioritization Template:

Brainstorming provides a free and open environment that encourages everyone within a team to participate in the creative thinking process that leads to problem solving. Prioritizing volume over value, out-of-the-box ideas are welcome and built upon, and all participants are encouraged to collaborate, helping each other develop a rich amount of creative solutions.

Problem Statement:

Finding rental houses is often time-consuming because information is scattered across different platforms. Students, families, and working professionals face difficulties in comparing properties, contacting owners, and identifying genuine listings.

Selected Project:

HouseHunt – A web-based house rental platform that allows users to search, view, and rent houses easily while enabling owners to list and manage their properties.

Step-1: Team Gathering, Collaboration and Select the Problem Statement



Brainstorm & idea prioritization

Use this template in your own brainstorming sessions so your team can unleash their imagination and start shaping concepts even if you're not sitting in the same room.

 10 minutes to prepare
 1 hour to collaborate
 2-8 people recommended

Before you collaborate

A little bit of preparation goes a long way with this session. Here's what you need to do to get going.

 10 minutes

A

Team gathering

Define who should participate in the session and send an invite. Share relevant information or pre-work ahead.

B

Set the goal

Think about the problem you'll be focusing on solving in the brainstorming session.

C

Learn how to use the facilitation tools

Use the Facilitation Superpowers to run a happy and productive session.

[Open article](#) →

1 Define your problem statement

People often struggle to find suitable rental houses because property information is scattered, making the search process time-consuming and inconvenient.

 5 minutes

PROBLEM

How can we simplify the process of finding suitable rental houses?

2

Key rules of brainstorming

To run a smooth and productive session

 Stay in topic.

 Encourage wild ideas.

 Defer judgment.

 Listen to others.

 Go for volume.

 If possible, be visual.

Brainstormed Ideas:

- Online House Rental Platform
- PG & Hostel Finder
- Property Recommendation System
- House Visit Booking
- Rental Agreement Generator

Idea Grouping:

User Features:

- Register/Login
- Search Houses
- View Property Details
- Chat with Owners
- Send Rental Offers

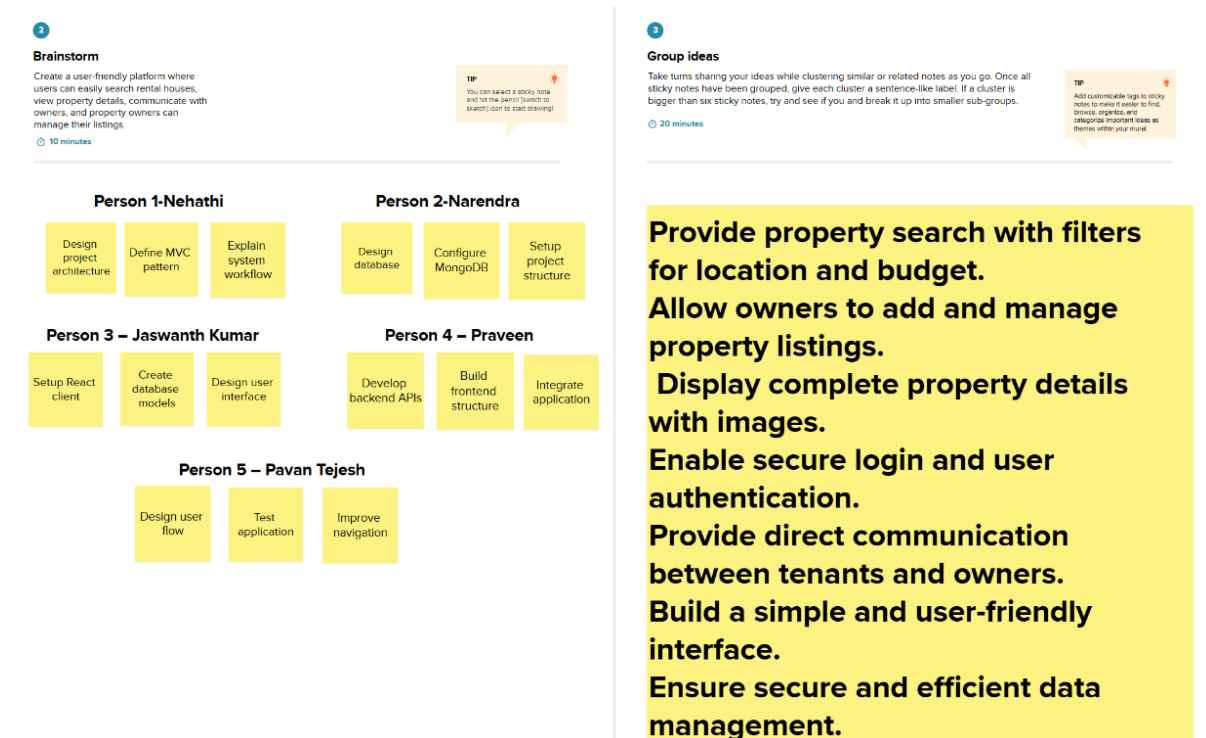
Owner Features:

- Add Property
- Update Property
- Delete Property
- Manage Listings

Admin Features:

- Monitor Users
- Manage Properties
- Handle Reports

Step-2: Brainstorm, Idea Listing and Grouping



Final Selected Idea: HouseHunt

Reason for Selection:

- Solves a real-life problem.
- Easy for users to find rental houses.
- Secure login and property management.
- Chat and offer features improve communication.
- Practical full-stack web application using React, Node.js, Express.js, and MongoDB.

Step-3: Idea Prioritization



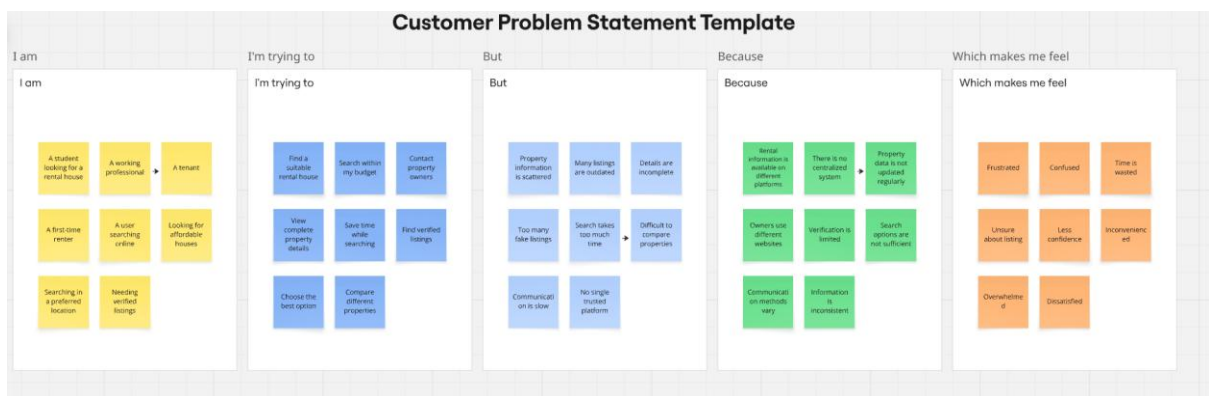
Ideation Phase

Define the Problem Statements

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Maximum Marks	2 Marks

Customer Problem Statement Template:

I am a tenant trying to find a suitable rental house, but property information is scattered and often outdated because there is no centralized platform, which makes the search process frustrating and time-consuming



Example:



Problem Statement (PS)	I am (Customer)	I'm trying to	But	Because	Which makes me feel
PS-1	I am a tenant looking for a suitable rental house	I'm trying to find a verified rental house within my budget and preferred location	But the property information is scattered, outdated, and difficult to compare	Because there is no single platform that provides complete and reliable rental information	Which makes me feel frustrated and wastes my time while searching for a suitable house

Ideation Phase Empathize & Discover

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Maximum Marks	4 Marks

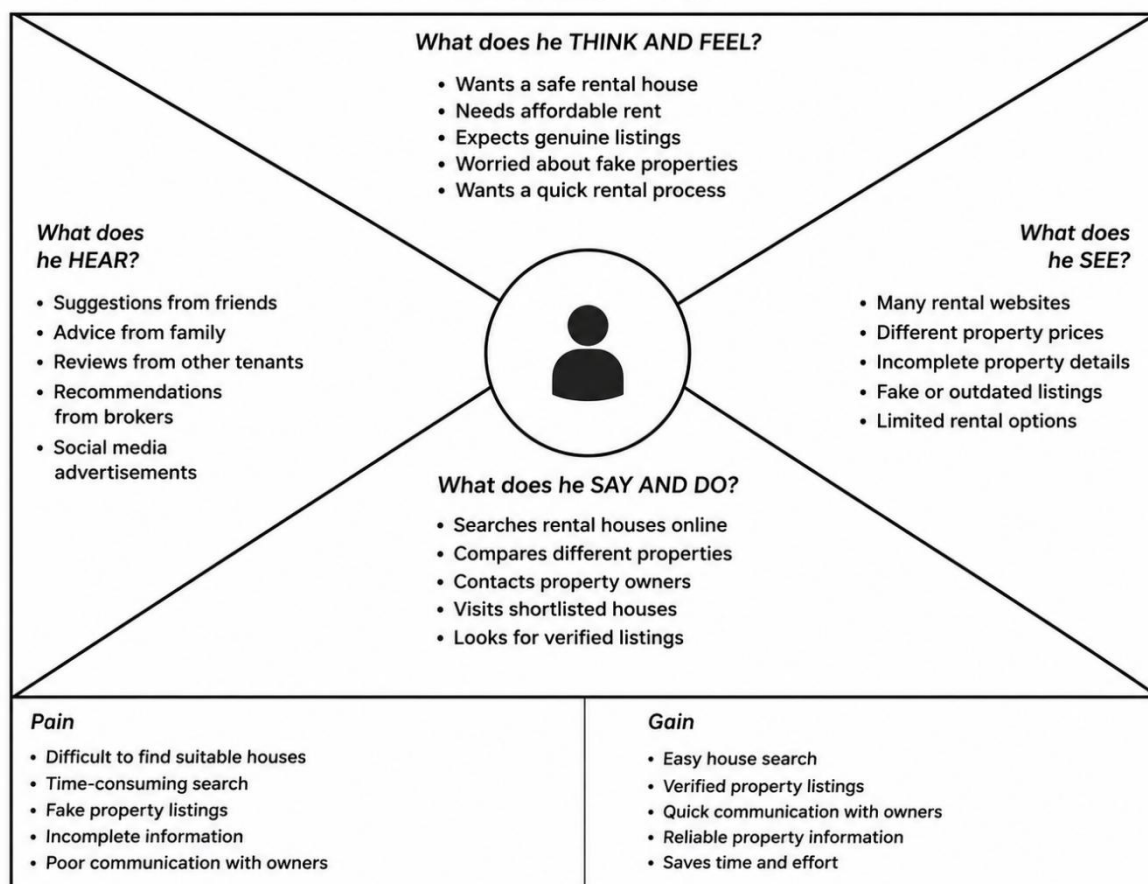
Empathy Map Canvas:

An empathy map is a simple, easy-to-digest visual that captures knowledge about a user's behaviours and attitudes.

It is a useful tool to help teams better understand their users.

Creating an effective solution requires understanding the true problem and the person who is experiencing it. The exercise of creating the map helps participants consider things from the user's perspective along with his or her goals and challenges.

Empathy Map



Project Design Phase
Problem – Solution Fit Template

Date	15 July 2026
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Project Name	HouseHunt – Smart House Rental Platform
Maximum Marks	2 Marks

Problem – Solution Fit Template:

The Problem–Solution Fit helps identify the challenges faced by people searching for rental houses and ensures that the proposed solution effectively addresses those challenges. In the HouseHunt project, the focus is on understanding users' needs and providing a simple, reliable, and user-friendly platform to make the house rental process easier and more efficient.

Purpose:

Understand the problems faced by tenants while searching for rental houses.

Identify the needs and expectations of property owners and tenants.

Design a simple and reliable platform that addresses real user problems.

Improve communication between tenants and property owners through a centralized system.

Provide an efficient house rental experience with verified property information and easy property management.

Template:

Define CS, fit into CL	1. CUSTOMER SEGMENT(S) CS • Students • Working professionals • Families • Property owners • House renters	6. CUSTOMER LIMITATIONS EG. BUDGET, DEVICES CL • Limited budget • Lack of verified information • Limited property choices • Time constraints • Difficulty contacting owners	5. AVAILABLE SOLUTIONS AS PROS & CONS • Property websites • Social media groups • Real estate brokers • Newspaper advertisements • Word-of-mouth recommendations	Explore AS, differentiate							
	2. PROBLEMS / PAINS + ITS FREQUENCY PR • Difficulty finding suitable rental houses • Scattered property information • Outdated or fake property listings • Time-consuming search process • Limited communication with property owners	9. PROBLEM ROOT / CAUSE RC • No centralized rental platform • Scattered property information • Unverified property listings • Poor communication between tenants and owners • Inconsistent property updates	7. BEHAVIOR + ITS INTENSITY BE • Search online for rental houses • Compare multiple properties • Contact property owners • Visit shortlisted houses • Check reviews and details		Focus on PR, tap into BE, understand RC						
Identify strong TR & EM	3. TRIGGERS TO ACT TR • Relocating to a new city • College admission • Job transfer • Need for a better house • Expiring rental agreement	10. YOUR SOLUTION SL HouseHunt is a web-based house rental platform that allows users to search rental houses, view verified property details, connect directly with property owners, and manage property listings through a simple and user-friendly interface.	8. CHANNELS of BEHAVIOR CH ONLINE • Rental websites • Social media • Search engines	Extract online & offline CH of BE							
	4. EMOTIONS BEFORE / AFTER EM <table><tr><td>Before</td><td>After</td></tr><tr><td>• Frustrated</td><td>• Satisfied</td></tr><tr><td>• Confused</td><td>• Confident</td></tr><tr><td>• Stressed</td><td>• Happy</td></tr><tr><td>• Uncertain</td><td>• Relaxed</td></tr></table>		Before		After	• Frustrated	• Satisfied	• Confused	• Confident	• Stressed	• Happy
Before	After										
• Frustrated	• Satisfied										
• Confused	• Confident										
• Stressed	• Happy										
• Uncertain	• Relaxed										

Project Design Phase Proposed Solution Template

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Maximum Marks	2 Marks

Proposed Solution Template:

Project team shall fill the following information in the proposed solution template.

S.No.	Parameter	Description
1.	Problem Statement (Problem to be solved)	Finding a suitable rental house is difficult because property information is scattered, outdated, and difficult to verify, making the search process time-consuming.
2.	Idea / Solution description	HouseHunt is a web-based house rental platform that helps users search rental houses, view property details, contact property owners,

		and manage property listings through a simple and user-friendly interface.
3.	Novelty / Uniqueness	The platform combines property search, verified listings, owner communication, and property management in a single application, providing a simple and organized rental experience.
4.	Social Impact / Customer Satisfaction	HouseHunt reduces the time and effort required to find rental houses, improves communication between tenants and owners, and provides reliable property information for a better user experience.
5.	Business Model (Revenue Model)	Revenue can be generated through premium property listings, featured advertisements, subscription plans for property owners, and promotional partnerships.
6.	Scalability of the Solution	The platform can be expanded by supporting multiple cities, adding advanced search filters, integrating online payment services, and providing mobile application support in the future.

Project Design Phase Solution Architecture

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Solution Architecture:

The HouseHunt solution architecture defines how different components of the application work together to simplify the house rental process. It connects the user interface, backend services, and database to provide a secure and efficient rental platform.

Its goals are to:

- Provide a simple and reliable platform for tenants and property owners.

- Define the interaction between the frontend, backend, and database for smooth application performance.
- Support features such as user authentication, property listing, property search, property management, and communication between users.
- Ensure secure data management, scalability, and efficient system performance for future enhancements.

Solution Architecture Diagram:

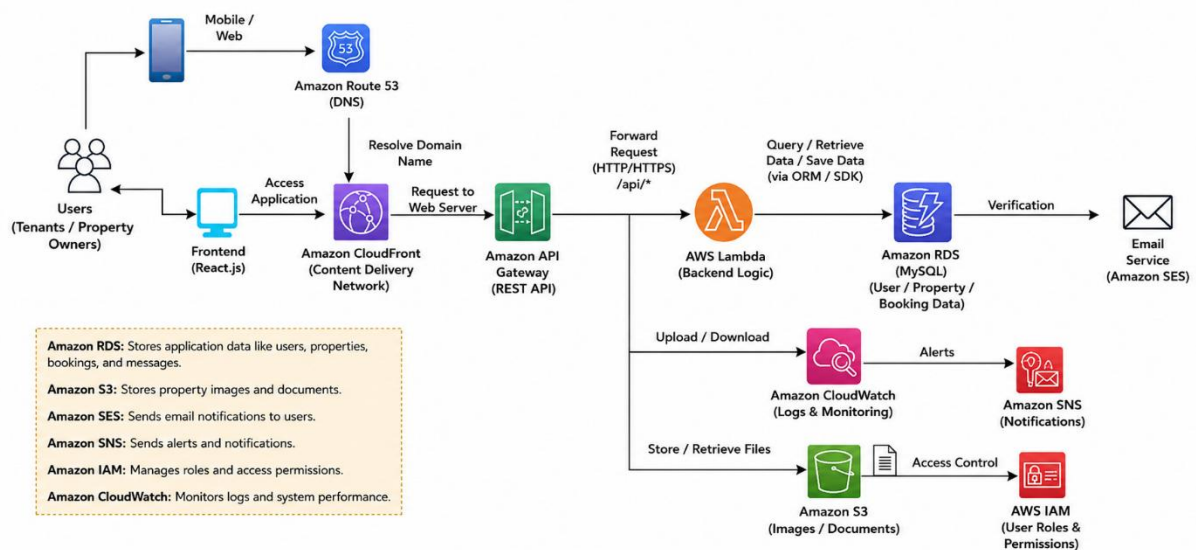


Figure 1: Architecture and data flow of the HouseHunt web application

User Acceptance Testing (UAT) Template

Date	15 July 2026
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Project Name	HouseHunt – Smart House Rental Platform
Maximum Marks	

Project Overview:

Project Name: HouseHunt – House Rental Platform

Project Description: HouseHunt is a web-based house rental application that helps users search for rental properties, view property details, communicate with property owners, and manage property listings through a simple and user-friendly interface.

Project Version: 1.0

Testing Period: 09 July 2026 to 14 July 2026

Testing Scope:

Features and Functionalities to be Tested

- User Registration
- User Login
- Property Listing
- Property Search
- Property Details
- Add Property
- Edit Property
- Delete Property
- Chat with Property Owner
- Rental Offer Management
- User Dashboard

User Stories / Requirements to be Tested

- Users should be able to register successfully.
- Users should be able to log in securely.
- Property owners should be able to add and manage listings.
- Users should be able to search and view properties.
- Users should be able to contact property owners.
- Users should be able to send rental offers.

Testing Environment:

URL/Location: <http://localhost:5173>

Credentials (if required): Username: testuser@gmail.com

Test Cases:

Test Case ID	Test Scenario	Test Steps	Expected Result	Actual Result	Pass/Fail
TC-001	User Registration	• Open Register • Enter valid details • Click Register	User account created successfully	Account created successfully	Pass

TC-002	User Login	• Open Login • Enter valid credentials • Click Login	User logged in successfully	User logged in successfully	Pass
TC-003	Search Property	• Enter location • Click Search	Matching properties displayed	Matching properties displayed	Pass
TC-004	View Property Details	• Select property • Open details page	Property details displayed	Property details displayed	Pass
TC-005	Add Property	• Login as Owner • Fill property details • Click Add Property	Property added successfully	Property added successfully	Pass
TC-006	Edit Property	• Select property • Update details • Save changes	Property updated successfully	Property updated successfully	Pass
TC-007	Delete Property	• Select property • Click Delete • Confirm	Property deleted successfully	Property deleted successfully	Pass
TC-008	Send Rental Offer	• Open property • Submit rental offer	Offer sent successfully	Offer sent successfully	Pass
TC-009	Chat with Owner	• Open Chat • Send message	Message sent successfully	Message sent successfully	Pass
TC-010	Logout	• Click Logout	User logged out successfully	User logged out successfully	Pass

Bug Tracking:

Bug ID	Bug Description	Steps to Reproduce	Severity	Status	Additional Feedback
BG-001	Property image loading delay	• Login • Open Property Details • View images	Low	Closed	Optimized image loading
BG-002	Invalid email accepted	• Open Register	Medium	Closed	Validation added

	during registration	• Enter invalid email • Click Register			
BG-003	Empty search displays all properties	• Leave search empty • Click Search	Low	Closed	Validation implemented
BG-004	Long property description affects layout	• Add long description • Open Property Details	Low	Closed	UI layout fixed

Sign-off:

Tester Name: Nehathi

Date: 15 July 2026

Signature: Nehathi

Notes:

- Ensure all HouseHunt features are tested successfully
- Execute both positive and negative test cases
- Record bugs with severity and status
- Verify all reported issues before deployment

Project Planning Phase

Project Planning Template (Product Backlog, Sprint Planning, Stories, Story points)

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Maximum Marks	5 Marks

Product Backlog, Sprint Schedule, and Estimation (4 Marks)

Use the below template to create product backlog and sprint schedule

Sprint	Functional Requirement (Epic)	User Story Number	User Story / Task	Story points
Sprint-1	User Authentication	USN-1	As a user, I can register for a HouseHunt account	2

Sprint	Functional Requirement (Epic)	User Story Number	User Story / Task	Story Points
Sprint-1	User Authentication	USN-2	As a user, I can log in securely	1
Sprint-1	Property Search	USN-3	As a user, I can search properties by location	2
Sprint-2	Property Management	USN-4	As an owner, I can add property details	3
Sprint-2	Property Management	USN-5	As an owner, I can edit and delete properties	2
Sprint-3	Property Details	USN-6	As a user, I can view complete property details	2
Sprint-3	Chat	USN-7	As a user, I can chat with property owners	3
Sprint-4	Offers	USN-8	As a user, I can send rental offers	2
Sprint-4	Profile	USN-9	As a user, I can update my profile	1
Sprint-4	Dashboard	USN-10	As an owner, I can manage listed properties	2

Project Tracker, Velocity & Burndown Chart: (4 Marks)

Sprint	Total Story Points	Duration	Sprint Start Date	Sprint End Date (Planned)	Story Points Completed / Planned
Sprint-1	User Authentication	USN-1	As a user, I can register for a HouseHunt account	2	High
Sprint-1	User Authentication	USN-2	As a user, I can log in securely	1	High
Sprint-1	Property Search	USN-3	As a user, I can search properties by location	2	High
Sprint-2	Property Management	USN-4	As an owner, I can add property details	3	High
Sprint-2	Property Management	USN-5	As an owner, I can edit and delete properties	2	Medium

Sprint	Total Story Points	Duration	Sprint Start Date	Sprint End Date (Planned)	Story Points Completed Planned E
Sprint-3	Property Details	USN-6	As a user, I can view complete property details	2	High
Sprint-3	Chat	USN-7	As a user, I can chat with property owners	3	Medium
Sprint-4	Offers	USN-8	As a user, I can send rental offers	2	Medium
Sprint-4	Profile	USN-9	As a user, I can update my profile	1	Low
Sprint-4	Dashboard	USN-10	As an owner, I can manage listed properties	2	High

Velocity:

Imagine we have a 10-day sprint duration, and the velocity of the team is 20 (points per sprint). Let's calculate the team's average velocity (AV) per iteration unit (story points per day)

$$AV = \frac{\text{sprint duration}}{\text{velocity}} = \frac{20}{10} = 2$$

Project Design Phase-II Data Flow Diagram & User Stories

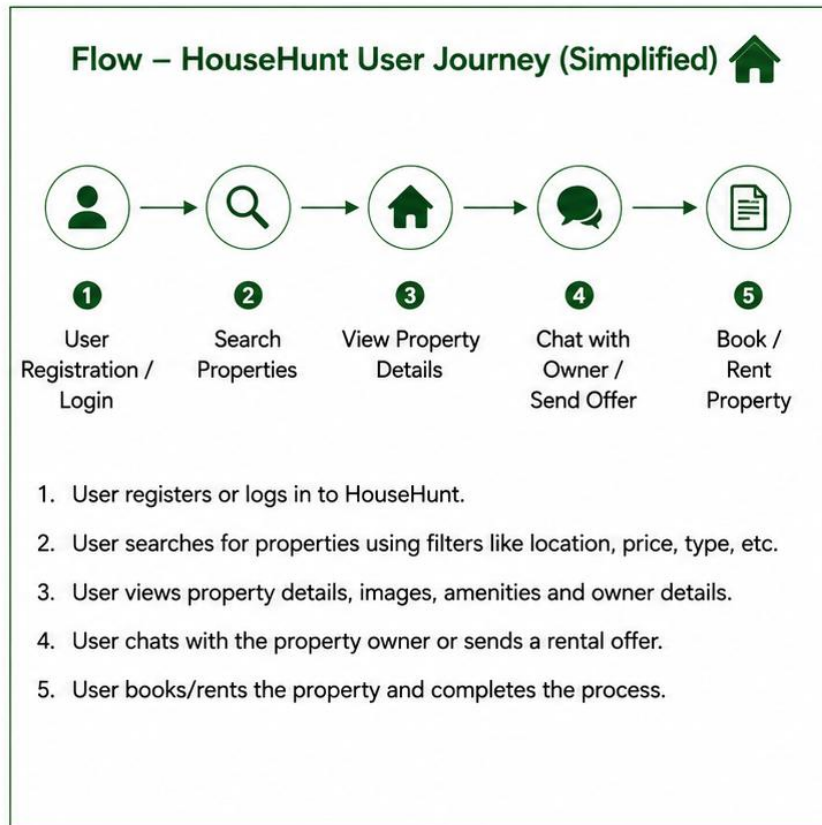
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Data Flow Diagrams:

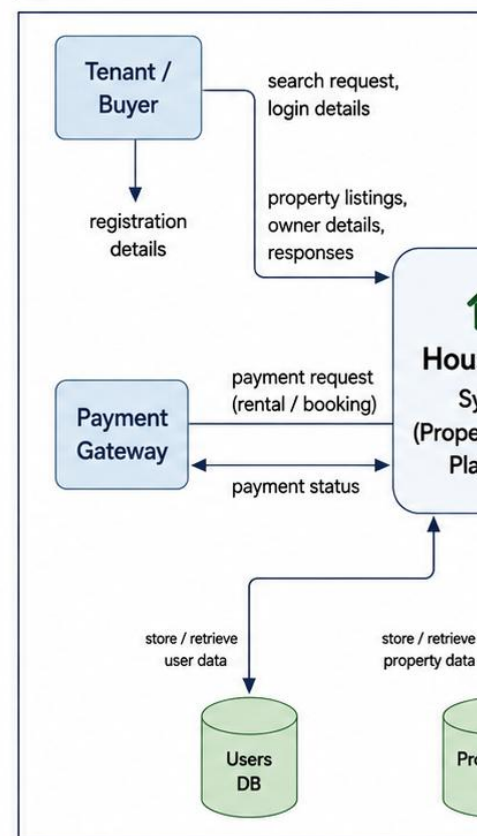
A Data Flow Diagram (DFD) is a traditional visual representation of the information flows within a system. A neat and clear DFD can depict the right amount of the system requirement graphically. It shows how data enters and leaves the system, what changes the information, and where data is stored.

Example: [\(Simplified\)](#)

Example: [\(HouseHunt\)](#)



Example: DFD Level 0 (HouseHunt – In



User Stories

Use the below template to list all the user stories for the product.

User Type	Functional Requirement (Epic)	User Story Number	User Story / Task	Acceptance
Customer (Mobile user)	Registration	USN-1	As a user, I can register for the application by entering my email, password, and confirming my password.	I can access the dashboard
		USN-2	As a user, I will receive confirmation email once I have registered for the application	I can receive email & click on the link
		USN-3	As a user, I can register for the application through Facebook	I can register the dashboard with Facebook Login

User Type	Functional Requirement (Epic)	User Story Number	User Story / Task	Acceptance
		USN-4	As a user, I can register for the application through Gmail	
	Login	USN-5	As a user, I can log into the application by entering email & password	
	Dashboard			
Customer (Web user)				
Customer Care Executive				
Administrator				

Project Design Phase-II
Solution Requirements (Functional & Non-functional)

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Functional Requirements:

Following are the functional requirements of the proposed solution.

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)
FR-1	User Registration	Registration through Form Registration through Gmail Registration through LinkedIn
FR-1	User Registration	Registration through Form User Login Password Validation

FR-2	Property Search	Search by Location Search by Budget Property Filters
FR-3	Property Management	Add Property Edit Property Delete Property View Property
FR-4	User Profile	View Profile Update Profile Change Password
FR-5	Property Details	View Images View Amenities Owner Information
FR-6	Chat & Communication	Chat with Property Owner Send Messages
FR-7	Rental Offers	Send Rental Offer View Offer Status
FR-8	Dashboard	Manage Listings Manage Offers Manage Profile

Non-functional Requirements:

Following are the non-functional requirements of the proposed solution.

FR No.	Non-Functional Requirement	Description
NFR-1	Usability	
NFR-1	Usability	Easy-to-use interface with simple navigation
NFR-2	Security	Secure authentication and protected user data
NFR-3	Reliability	Consistent and reliable system performance
NFR-4	Performance	Fast property search and quick response time
NFR-5	Availability	System available for users whenever required
NFR-6	Scalability	Supports increasing users and property listings

Project Design Phase-II Technology Stack (Architecture & Stack)

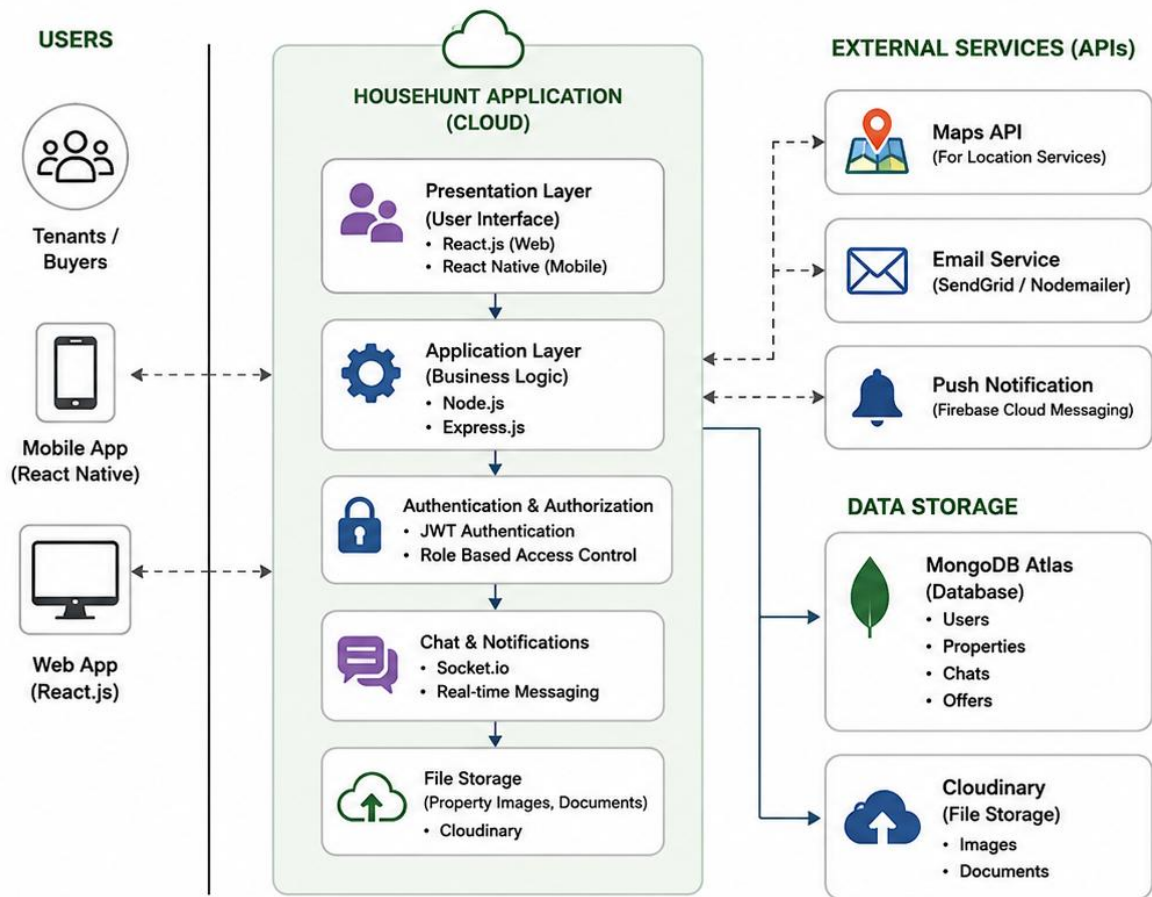
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Technical Architecture:

The Deliverable shall include the architectural diagram as below and the information as per the table1 & table 2

Example: Order processing during pandemics for offline mode

HouseHunt – Technical Architecture



Guide



Table-1 : Components & Technologies:

S.No	Component	Description	Tech
1	User Interface	Web application for tenants and owners	React
2	Application Logic-1	Authentication & User Management	Node
3	Application Logic-2	Property Management	Express
4	Application Logic-3	Chat & Rental Offers	Node
5	Database	Store users, properties, chats and offers	Mongo
6	Cloud Database	Database hosting	Mongo
7	File Storage	Property images	Cloud
8	External API-1	Maps / Location	Google
9	External API-2	Email notifications	Node
10	Machine Learning Model	Not Applicable	N/A
11	Infrastructure	Application deployment	Verac

Table-2: Application Characteristics:

S.No	Characteristics	Description	Tech
1	Open-Source Frameworks	Frameworks used	React
2	Security Implementations	Authentication and data protection	JWT,
3	Scalable Architecture	Three-tier architecture	React
4	Availability	Cloud deployment	Verac
5	Performance	Fast API responses and optimized UI	REST

